

CHAPTER C19

SOIL–STRUCTURE INTERACTION FOR SEISMIC DESIGN

C19.1 GENERAL

In an earthquake, the shaking is transmitted up through the structure from the geologic media underlying and surrounding the foundation. The response of a structure to earthquake shaking is affected by interactions among three linked systems: the structure, the foundation, and the geologic media underlying and surrounding the foundation. The analysis procedures in Chapters 12 and 15 idealize the response of the structure by applying forces to the structure, which is typically assumed to have a fixed base at the foundation–soil interface. In some cases, the flexibility of the foundation elements and underlying soils is included in the analysis model. The forces that are applied to the structure are devised based on parameters representing free-field ground motions. The term “free-field” refers to motions not affected by structural vibrations or the foundation characteristics of the specific structure and represents the condition for which the design spectrum is derived using the procedures given in Chapter 11 and Chapter 21. In most cases, however, the motions at the foundation that are imparted to the structure are different from the free-field motions. This difference is caused by the effects of the interaction of the structure and the geologic media. A seismic soil–structure interaction (SSI) analysis evaluates the collective response of these systems to a specified free-field ground motion.

SSI effects are absent for the theoretical condition of rigid geologic media, which is typical of analytical models of structures. Accordingly, SSI effects reflect the differences between the actual response of the structure and the response for the theoretical, rigid base condition. Visualized within this context, two following SSI effects can significantly affect the response of structures:

1. **Foundation Deformations.** Flexural, axial, and shear deformations of foundation elements occur as a result of loads applied by the superstructure and the supporting geologic media. Additionally, the underlying geologic media deforms because of loads from the foundations. Such deformations represent the seismic demand for which foundation components should be designed. These deformations can also significantly affect the overall system behavior, especially with respect to damping.
2. **Inertial SSI Effects.** Inertia developed in a vibrating structure gives rise to base shear, moment, and torsional excitation, and these loads in turn cause displacements and rotations of the foundation relative to the free-field displacement. These relative displacements and rotations are only possible because of flexibility in the soil–foundation system, which can significantly contribute to the overall structural flexibility in some cases. Moreover, the relative foundation free-field motions give rise to energy dissipation via radiation damping (i.e., damping associated with wave propagation into the ground away from the

foundation, which acts as the wave source) and hysteretic soil damping, and this energy dissipation can significantly affect the overall damping of the soil–foundation–structure system. Because these effects are rooted in the structural inertia, they are referred to as inertial interaction effects.

3. **Kinematic SSI Effects.** Kinematic SSI results from the presence of foundation elements on or in soil that are much stiffer than the surrounding soil. This difference in stiffness causes foundation motions to deviate from free-field motion as a result of base slab averaging and embedment effects.

Chapter 19 addresses both types of SSI effects. Procedures for calculating kinematic and inertial SSI effects were taken from recommendations in NIST GCR 12-917-21 (NIST 2012). Further discussion of SSI effects can be found in this NIST document and some of the references cited therein.

Substantial revisions have been made to Chapter 19 in this edition of ASCE 7. They include (1) the introduction of formulas for the stiffness and damping of rectangular foundations, (2) revisions to the formulas for the reduction of base shear caused by SSI, (3) reformulation of the effective damping ratio of the SSI system, (4) introduction of an effective period lengthening ratio, which appears in the formula for the effective damping ratio of the SSI system, and which depends on the expected structural ductility demand, and (5) the introduction of kinematic SSI provisions. Most of these revisions come from the NIST GCR 12-917-21 (NIST 2012) report on SSI. However, the basic model of the inertial SSI system has remained the same since SSI provisions were first introduced in the ATC 3-06 report (ATC 1978).

The first effect, foundation deformation, is addressed by explicitly requiring the design professional to incorporate the deformation characteristics of the foundation into their analysis model. Including foundation deformations is essential for understanding soil–structure interaction (SSI). Therefore, the flexibility of the foundation must be modeled to capture translational and rotational movement of the structure at the soil–foundation interface.

For the linear procedures, this requirement to model the flexibility of the foundation and soil means that springs should be placed in the model to approximate the effective linear stiffness of the deformations of the underlying geologic media and the foundation elements. This could be done by placing isolated spring elements under the columns and walls, by explicitly modeling the foundation elements and geologic media in the mathematical model, or some combination of the two. For the response history procedure, this would mean that in addition to the stiffness of the subsurface media and foundation elements, the nonlinear parameters of those materials would be incorporated into the analytical model. Because of the uncertainty in estimating the stiffness and deformation capacity of geologic media, upper and lower bound estimates of the properties should

be used and the condition that produces the more conservative change in response parameters from a fixed-base structure must be used.

Inertial interaction effects are addressed through the consideration of foundation damping. Inertial interaction in structures tends to be important for stiff structural systems such as shear walls and braced frames, particularly where the foundation soil is relatively soft. The provisions provide a method for estimating radiation damping and soil hysteretic damping.

The two main kinematic interaction effects are included in these provisions: base slab averaging and embedment effects. The kinematic interaction effects cause the motion input into the structure to be different from the free-field motions. The provisions provide a means by which a free-field, site-specific response spectrum can be modified to account for these kinematic interaction effects to produce a foundation-input spectrum.

Site classes A and B are excluded from Chapter 19 because the dynamic interaction between structures and rock is minimal based on theory. Furthermore, there are no empirical data to indicate otherwise.

Section 19.1.1 prohibits using the cap of S_s included in Section 12.8.1.3 because of the belief that structures meeting the requirements of that section have performed satisfactorily in past earthquakes, partially because of SSI effects. Taking advantage of that predetermined cap on S_s and then subsequently reducing the base shear caused by SSI effects may therefore amount to double-counting the SSI effects.

C19.2 SSI ADJUSTED STRUCTURAL DEMANDS

When the equivalent lateral force procedure is used, the equivalent lateral force is computed using the period of the flexible base structure and is modified for the SSI system damping. For the modal analysis procedure, a response spectrum, which has been modified for the SSI system damping and then divided by (R/I_e) , is input into the mathematical model. The lower bound limit on the design base shear based on the equivalent lateral force procedures per Section 12.9.1.4 still applies, but the equivalent lateral force base shear modified to account for SSI effects replaces the base shear for the fixed-base case.

For both the equivalent lateral force and response spectrum procedures, the total reduction caused by SSI effects is limited to a percentage of the base shear determined in accordance with Section 12.8.1, which varies based on the R factor. This limitation on potential reductions caused by SSI reflects the limited understanding of how the effects of SSI interact with the R factor. All of the SSI effects presented herein are based on theoretical linear elastic models of the structure and geologic media. That is why reductions of 30% are permitted for $R=3$ or less. It is felt that those systems exhibit limited inelastic response and therefore, a larger reduction in the design force caused by SSI should be permitted. For higher R factor systems, where significant damping caused by structural yielding is expected, the contribution of foundation damping is assumed to have little effect on the reduction of the response. Some reduction is permitted because of (1) an assumed period lengthening resulting from the incorporation of base flexibility, (2) potential reduction in mass participation in the fundamental mode because two additional degrees of freedom are present caused by translation and rotation of the base, and (3) limited foundation damping interacting with the structural damping.

Reductions to the response spectrum caused by the SSI system damping and kinematic SSI effects are for the elastic 5% damped response spectrum typically provided to characterize free-field motion. In addition, studies have indicated that there is a fair

amount of uncertainty in the amount of kinematic SSI when measured reductions between the free-field motion and the foundation input motion are compared with the theoretical models (Stewart 2000).

Reductions for kinematic SSI effects are not permitted for the equivalent lateral force and modal response spectrum procedures. The equations for predicting the kinematic SSI effects are based on modifications to the linear elastic response spectrum. Studies have not been performed to verify if they are similarly valid for inelastic response spectra, on which the R factor procedures are based. Additionally, the amount of the reduction for kinematic SSI effects is dependent on the period of the structure, with the greatest modifications occurring in the short period range. Because the fundamental periods of most structures lengthen as they yield, what would potentially be a significant reduction at the initial elastic period may become a smaller reduction as the structure yields. Without an understanding of how the period may lengthen in the equivalent lateral force or modal response spectrum procedures, there is a potential for a user to overestimate the reduction in the response parameters caused by kinematic SSI effects. Thus, their use is not permitted.

All types of SSI effects are permitted to be considered in a response history analysis per Chapter 16. If SSI effects are considered, the site-specific response spectrum should be used as the target to which the acceleration histories are scaled. The requirement to use a site-specific response spectrum was placed in the provisions because of the belief that it provided a more realistic definition of the earthquake shaking than is provided by the design response spectrum and MCE_R response spectrum in accordance with Sections 11.4.6 and 11.4.7. A more realistic spectrum was required for proper consideration of SSI effects, particularly kinematic SSI effects. The design response spectrum and MCE_R response spectrum, in accordance with Sections 11.4.6 and 11.4.7, use predetermined factors to modify the probabilistic or deterministic response spectrum for the soil conditions. These factors are sufficient for most design situations. However, if SSI effects are to be considered and the response spectrum modified accordingly, then more accurate representations of how the underlying geologic media alter the spectral ordinates should be included before the spectrum is modified because of the SSI effects.

A site-specific response spectrum that includes the effects of SSI can be developed with explicit consideration of SSI effects by modifying the spectrum developed for free-field motions through the use of the provisions in Sections 19.3 and 19.4. If the foundation damping is not specifically modeled in the analytical model of the structure, the input response spectrum can include the effects of foundation damping. Typically, the base slab averaging effect is not explicitly modeled in the development of a site-specific response spectrum and the provisions in Section 19.4.1 are used to modify the free-field, site-specific response spectrum to obtain the foundation input spectrum. Embedment effects can be modeled directly by developing the site-specific spectrum at the foundation base level, as opposed to the ground surface. Alternatively, the site-specific spectrum for the free field can be developed at the ground level and the provisions of Section 19.4.2 can be used to adjust it to the depth corresponding to the base of foundation.

The limitations on the reductions from the site-specific, free-field spectrum to the foundation input spectrum are based on several factors. The first is the scatter between measured ratios of foundation input motion to free-field motion versus the ratios from theoretical models (Stewart 2000). The second is the inherent variability of the properties of the underlying geologic media over the footprint of the structure. Whereas there is a

requirement to bound the flexibility of the soil and foundation springs, there are no corresponding bounding requirements applied to the geologic media parameters used to compute the foundation damping and kinematic SSI. The last factor is the aforementioned lack of research into the interaction between SSI effects and yielding structures. Some studies have shown that there are reductions for most cases of SSI when coupled with an R factor based approach (Jaremrprasert et al. 2013).

A limitation was placed on the maximum reduction for an SSI modified site-specific response spectrum with respect to the response spectrum developed based on the USGS ground-motion parameters and the site coefficients. This limitation is caused by similar concerns expressed in Section C21.3 regarding the site-specific hazard studies generating unreasonably low response spectra. There is a similar concern that combining SSI effects with site-specific ground motions could significantly reduce the seismic demand from that based on the USGS ground-motion parameters and the site coefficients. However, it was recognized that these modifications are real and the limit could be relaxed, but not eliminated, if there were (1) adequate peer review of the site-specific seismic hazard analysis and the methods used to determine the reductions attributable to SSI effects and (2) approval of the jurisdictional authority.

Peer review would include, but not be limited, to the following:

1. Development of the site-specific response spectrum used to scale the ground motions;
2. Determination of foundation stiffness and damping, including the properties of the underlying subsurface media used in the determination;
3. Confirmation that the base slab and first slab above the base are sufficiently rigid to allow base slab averaging to occur, including verification that the base slab is detailed to act as a diaphragm; and
4. Assumptions used in the development of the soil and radiation damping ratios.

The SSI effects can be used in a response history analysis per Chapter 16. Two options for the modeling of the SSI are as follows:

1. Create a nonlinear finite element (FE) model of the structure, foundation, and geologic media. The mesh for the geologic media should extend to an appropriate depth and horizontal distance away from the foundation with transmitting boundaries along the sides to absorb outgoing seismic waves generated by the foundation. The motion should be input at the base of the FE model and should propagate upward as shear waves. The free-field response spectrum can be reduced for kinematic SSI only per the provisions in Section 19.4, but embedment effects would not be allowed in the reduction because the waves propagating up from the depth of the foundation to the surface would automatically include kinematic effects of embedment.
2. Create a nonlinear finite element model of the structure and foundation, with springs and dashpots attached to the perimeter walls and base of the foundation to account for the soil–foundation interaction. Guidance on the development of dashpots can be found in NIST GCR 12-917-21 (NIST 2012). The free-field response spectrum can be reduced for kinematic SSI per Section 19.4, but embedment effects may or may not be allowed in the reduction depending on whether or not (i) the motion is allowed to vary with depth along the embedded portion of the

foundation, and (ii) the free-field motion used as input motion is defined at the ground surface or at the bottom of the basement. The dashpots would account for the radiation and hysteretic damping of the geologic media, either per Section 19.3 or more detailed formulations.

C19.3 FOUNDATION DAMPING

The procedures in Section 19.3 are used to estimate an SSI system damping ratio, β_0 , based on the underlying geologic media and interaction of the structure and its foundation with this geologic media. There are two main components that contribute to foundation damping: soil hysteretic damping and radiation damping. The provisions in this section provide simplified ways to approximate these effects. However, they are complex phenomena and there are considerably more detailed methods to predict their effects on structures. The majority of the provisions in this section are based on material in NIST GCR 12-917-21 (NIST 2012). Detailed explanations of the background of these provisions, supplemental references, and more sophisticated methods for predicting radiation damping can be found in that report. However, those references do not provide the derivation of the effective period lengthening ratio, $(\tilde{T}/T)_{\text{eff}}$ given by Eq. (19.3-2). This ratio appears in the equation for β_0 (Eq. 19.3-1), and it is derived from the total displacement of the mass of the SSI oscillator model resulting from a horizontal force applied to the mass. A component of this displacement is the displacement of the mass relative to its base, and it is equal to the ductility demand, μ , times the elastic displacement of the mass relative to the base. The other components of the total displacement arise from displacement of the translational foundation spring (K_y or K_r) and the translation resulting from the rotational foundation spring (K_{xx} or K_{rr}). The period lengthening ratio, $(\tilde{T}/T)$ appearing in Eq. (19.3-2) is derived in the same manner assuming that $\mu = 1$.

Radiation damping refers to energy dissipation from wave propagation away from the vibrating foundation. As the ground shaking is transmitted into the structure's foundation, the structure itself begins to translate and rock. The motion of the foundation relative to the free-field motion creates waves in the geologic media, which can act to counter the waves being transmitted through the geologic media caused by the earthquake shaking. The interference is dependent on the stiffness of the geologic media and the structure, the size of the foundation, type of underlying geologic media, and period of the structure. The equations for radiation damping in Section 19.3.3 were taken from NIST GCR 12-917-21 (NIST 2012); details of the derivation are found in Givens (2013).

In Section 19.3.3, the equations for K_y and K_{xx} , for rectangular foundations, and the associated damping ratios, β_y and β_{xx} , come from Pais and Kausel (1988) and are listed in Table 2-2a and Table 2-3a in the NIST report. The corresponding static stiffness equations for circular foundations in Section 19.3.4 were taken from Veletsos and Verbic (1973); the other equations appearing in Section 19.3.4 were adapted from equations in the NIST report. The foundation stiffness and damping equations in these two sections apply to surface foundations. The reasons for excluding embedment effects are explained in the third paragraph from the end of this subsection.

Soil hysteretic damping occurs because of shearing within the soil and at the soil–foundation interface. Values of the equivalent viscous damping ratio, β_s , to model the hysteretic damping can be obtained from site response analysis or Table 19.3-3.

Foundation damping effects, modeled by β_f , tend to be important for stiff structural systems such as shear walls and

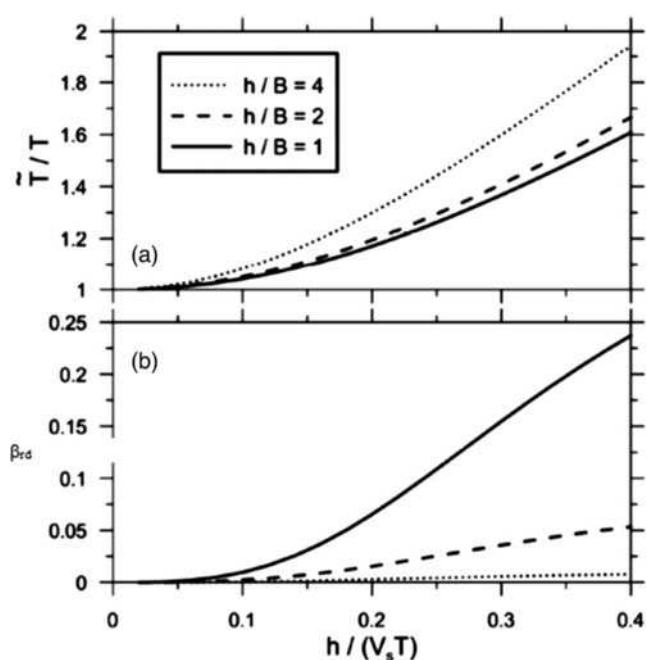


FIGURE C19.3-1 Example of Radiation Damping

Source: NIST 2012.

braced frames, particularly where they are supported on relatively soft soil sites, such as Site Classes D and E. This effect is determined by taking the ratio of the fundamental period of the structure, including the flexibility of the foundation and underlying subsurface media (flexible-base model) and the fundamental period of the structure assuming infinite rigidity of the foundation and underlying subsurface media (fixed-base model). Analytically, this ratio can be determined by computing the period of the structure with the foundation/soil springs in the model and then replacing those springs with rigid support.

Fig. C19.3-1 illustrates the effect of the period ratio, $\tilde{T}/T$, on the radiation damping, β_{rs} , which typically accounts for most of the foundation damping. $\tilde{T}/T$ is the ratio of the fundamental period of the SSI system to the period of the fixed-base structure. The figure shows that for structures with larger height, h , to foundation half-width, B , aspect ratios, the effects of foundation damping become less. In this figure, the aspect ratio of the foundation is assumed to be square.

These inertial interaction effects are influenced considerably by the shear modulus of the underlying subgrade, specifically the modulus that coincides with the seismic shaking being considered. As noted in the standard, shear modulus G can be evaluated from small-strain shear wave velocity as $G = (G/G_o)G_o = (G/G_o)\gamma v_{so}^2/g$ (all terms defined in the standard). Shear wave velocity, v_{so} , should be evaluated as the average small-strain shear wave velocity within the effective depth of influence below the foundation. The effective depth should be taken as half the lesser dimension of the foundation, which in the provisions is defined as B . Methods for measuring v_{so} (preferred) or estimating it from other soil properties are summarized elsewhere (e.g., Kramer 1996).

The radiation damping procedure is conservative and underestimates the foundation damping for shaking in the long direction where the foundation aspect ratios exceed 2:1 but could be potentially unconservative where wall and frame elements are close enough so that waves emanating from distinct foundation components destructively interfere with each other across the

period range of interest. That is why the limit of spacing of the vertical lateral force-resisting elements is imposed on the use of these provisions.

For structures supported on footings, the formulas for radiation damping can generally be used with B and L calculated using the footprint dimensions of the entire structure, provided that the footings are interconnected with grade beams and/or a sufficiently rigid slab on grade. An exception can occur for structures with both shear walls and frames, for which the rotation of the foundation beneath the wall may be independent of that for the foundation beneath the column (this type is referred to as weak rotational coupling). In such cases, B and L are often best calculated using the dimensions of the wall footing. Very stiff foundations like structural mats, which provide strong rotational coupling, are best described using B and L values that reflect the full foundation dimension. Regardless of the degree of rotational coupling, B and L should be calculated using the full foundation dimension if foundation elements are interconnected or continuous. Further discussion can be found in FEMA 440 (FEMA 2005) and NIST GCR 12-917-21 (NIST 2012).

The radiation damping provisions conservatively exclude the effects of embedment. Embedment typically increases the amount of radiation damping if the basement or below-grade foundation stays in contact with the soil on all sides. Because there is typically some gapping between the soil and the sides of the basement or foundation, these embedment effects may be less than the models predict. There are some additional issues with the procedures for embedded foundations. For the case where the embedment is significant but the soils along the sides are much more flexible than the bearing soils, a high impedance contrast between the first two layers is recognized as a potential problem regardless of the embedment. The NIST GCR 12-917-21 (NIST 2012) report therefore recommends ignoring the additional contributions caused by embedment but still using the soil properties derived below the embedded base.

The equations in Sections 19.3.3 and 19.3.4 are for shallow foundations. This is not to say that radiation damping does not occur with deep (pile or caisson) foundation systems, but the phenomenon is more complex. Soil layering and group effects are important, and there are the issues of the possible contributions of the bottom structural slab and pile caps. Because the provisions are based on the impedance produced by a rigid plate in soil, these items cannot be easily taken into account. Therefore, more detailed modeling of the soil and the embedded foundations is required to determine the foundation impedances. The provisions permit such modeling but do not provide specific guidance for it. Guidance can be found for example in NIST GCR 12-917-21 (NIST 2012) and its references.

Soil hysteretic damping occurs as seismic waves propagate through the subsurface media and reach the base of the structure, and it can have an effect on the overall system damping when the soil strains are high. Table 19.3-3 in the provisions was derived based on relationships found in EPRI (1993) and Vucetic and Dobry (1991) that relate the ratio between G/G_o to cyclic shear strain in the soil, and then to soil damping. The values in the table are based on conservative assumptions about overburden pressures on granular soils and plasticity index of clayey soils. This simplified approach does not preclude the geotechnical engineer from providing more detailed estimates of soil damping. However, the cap on reductions in the seismic demand are typically reached at around an additional 5% hysteretic damping ratio (10% total damping ratio), and further reductions would require peer review.

C19.4 KINEMATIC SSI EFFECTS

Kinematic SSI effects are broadly defined as the difference between the ground motion measured in a free-field condition and the motion which would be measured at the structure's foundation, assuming that it and the structure were massless (i.e., inertial SSI was absent). The differences between free-field and foundation input motions are caused by the characteristics of the structure foundation, exclusive of the soil and radiation damping effects in the preceding section. There are two main types of kinematic interaction effects: base slab averaging and embedment. The provisions provide simplified methods for capturing these effects. The basis for the provisions and additional background material can be found in FEMA 440 (FEMA 2005) and NIST GCR 12-917-21 (NIST 2012).

FEMA 440 (FEMA 2005) specifically recommends against applying these provisions to very soft soil sites such as E and F. These provisions allow kinematic SSI for Site Class E but retain the prohibition for Site Class F. That is not to say that kinematic interaction effects are not present at Site Class F sites, but that these specific provisions should not be used; rather, more detailed site-specific assessments are permitted to be used to determine the possible modifications at those sites.

In addition to the prescriptive methods contained in the standard, there are also provisions that allow for direct computation of the transfer function of the free-field motion to a foundation input motion caused by base slab averaging or embedment. Guidance on how to develop these transfer functions can be found in NIST GCR 12-917-21 (NIST 2012) and the references contained therein.

C19.4.1 Base Slab Averaging. Base slab averaging refers to the filtering of high-frequency portions of the ground shaking caused by the incongruence of motion over the base. For this filtering to occur, the base of the structure must be rigid or semirigid with respect to the vertical lateral force-resisting elements and the underlying soil. If the motions are out of phase from one end of the foundation to the other and the foundation is sufficiently rigid, then the motion on the foundation would be different from the ground motion at either end. The ground motions at any point under the structure are not in phase with ground motions at other points along the base of the structure. That incongruence leads to interference over the base of the structure, which translates into the motions imparted to the foundation, which are different from the ground motions. Typically, this phenomenon results in a filtering out of short-period motions, which is why the reduction effect is much more pronounced in structures with short fundamental periods, as illustrated in Fig. C19.4-1.

Fig. C19.4-1 illustrates the increase in reduction as the base area parameter, b_e , increases. This parameter is computed as the square root of the foundation area. Therefore, for larger foundations, base slab averaging effects are more significant.

For base slab averaging effects to occur, foundation components must be interconnected with grade beams or a concrete slab that is sufficiently stiff to permit the base to move as a unit and allow this filtering effect to occur. That is why requirements are placed on the rigidity of the foundation diaphragm relative to the vertical lateral force-resisting elements at the first story. Additionally, requirements are placed on the floor diaphragm or roof diaphragm, in the case of a one-story structure needing to be stiff in order for this filtering of ground motion to occur. FEMA 440 (FEMA 2005) indicates that there is a lack of data regarding this effect when either the base slab is not interconnected or the floor diaphragms are flexible. It is postulated that reductions between the ground motion and the foundation input motion may still occur. Because cases like this have not been studied in FEMA 440 (FEMA 2005) and NIST GCR

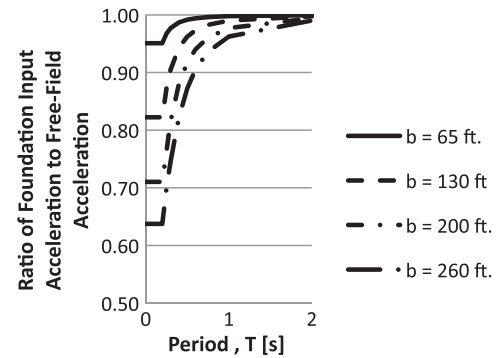


FIGURE C19.4-1 Example of Base Slab Averaging Response Spectra Ratios

12-917-21 (NIST 2012) explicitly, the requirements for foundation connectivity and stiff or rigid diaphragms above the foundation have been incorporated into the provisions.

The underlying models have only been studied up to an effective base size of 260 ft (79.2 m), which is why that limitation has been placed on Eq. (19.4-4). FEMA 440 (FEMA 2005) postulates that this effect is likely to still occur for larger base areas, but there has not been sufficient study to compare the underlying equations to data at larger effective base sizes.

Also, because the reduction can become quite significant and because studies of these phenomena have indicated variability between the theoretically predicted modifications and actual measured modifications (Stewart et al. 1999, Stewart 2000), a 0.75 factor is applied to the equations that are found in NIST GCR 12-917-21 (NIST 2012) to provide an upper bound estimate of the reduction factors with respect to the theoretical models. This is why the equations differ from those found in FEMA 440 (FEMA 2005).

Lastly, the method has not been rigorously studied for structures on piles (NIST 2012); however, it is considered reasonable to extend the application to pile-supported structures in which the pile caps are in contact with the soil and are laterally connected to one another. Another justification is that some of the empirical data for kinematic SSI come from pile-supported structures.

C19.4.2 Embedment. The kinematic interaction effects caused by embedment occur because the seismic motions vary with depth below the ground surface. It is common for these effects to be directly considered in a site-specific response spectrum by generating response spectra and acceleration histories at the embedded base of the structure instead of the ground surface. If that is not done, then these effects can be accounted for using the provisions in this section. However, these provisions should not be used if the response spectrum has already been developed at the embedded base of the structure. The embedment effect model was largely based on studies of structures with basements. The provisions can also be applied to structures with embedded foundations without basements where the foundation is laterally connected at the plane taken as the embedment depth. However, the provisions are not applicable to embedded individual spread footings.

As with base slab averaging, the reduction can become quite significant, and studies of these phenomena have indicated variability between the theoretically predicted modifications and actual measured modifications (Stewart et al. 1999). Again, a 0.75 factor is applied to the equations found in NIST GCR 12-917-21 (NIST 2012) to provide a slightly conservative estimate of the reductions with respect to the theoretical models. This is why the equations differ from those found in FEMA 440 (FEMA

2005) and NIST GCR 12-917-21 (NIST 2012). Additionally, the underlying models upon which the provisions are based have only been validated in NIST GCR 12-917-21 (NIST 2012) up to an effective embedment depth of approximately 20 ft (6.096 m), which is why a depth limitation has been placed on Eq. (19.2-4).

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